**To:** The Centers for Medicare & Medicaid Services (CMS) and the Assistant Secretary for Technology Policy (ASTP)

**Submitted by:** Teresa Younkin, MSHI | teresayounkin.com

AI Strategist | Health Informaticist | National Speaker: AI Governance and Data Standards

**Response to**: CMS RFI (CMS-0042-NC): Health Technology Ecosystem

# I. Introduction

After a doctor's appointment, as we filled up my mother's weekly medication box I watched her confusion when medications are removed from her medication list but continue arriving via auto-refill. This disconnect between health IT systems and human reality represents more than medication confusion—it's one symptom of a larger crisis: the massive coordination gap between what medical providers document and what family caregivers actually manage daily.

As a health informaticist and AI strategist with two decades of experience, I approach technology like a surgeon approaches sutures—stitching people and processes together with ones and zeroes to heal systemic gaps. My perspective is anchored in lived experience—from faxing records for my late husband's complex care to navigating today's fragmented digital landscape across multiple family members.

Through Mosaic Life Tech, I support organizations like the National Council for Prescription Drug Programs (NCPDP) in understanding the impact of AI with governance frameworks grounded in real-world impact and trust. My expertise spans SMART on FHIR applications, HL7 standards, and value-based care principles, but what drives my work is understanding the invisible health data that actually drives outcomes: caregiver observations of daily functioning, transportation barriers, family communication patterns, medication adherence challenges, and the countless coordination tasks that fall to families when clinical systems don't talk to each other.

AI's greatest opportunity lies in making this invisible caregiving work visible and actionable within health systems. The approach to this RFI committed to ensuring digital health ecosystems don't just innovate—they recognize and integrate the essential coordination work that family caregivers perform when medical providers' systems fail to connect.

# II. Patient and Consumer Responses

## Patient Access to Comprehensive Health Information

*This directly addresses RFI questions PC-1, PC-2, PC-8, PC-9, and PC-12, which focus on health management applications, consolidated health information access, valuable health data sources, Blue Button API expansion, and operational health data use cases.*

### Barriers

Patient and caregiver access to complete and understandable medical information remains fragmented across systems, portals, and formats. Clinical notes—when available—are often delayed, heavily templated, or full of clinical jargon that bears little resemblance to what was communicated during the visit. This misalignment undermines trust and makes it difficult for patients and caregivers to act on the information in meaningful ways.

Current EHRs are structured for billing and clinical abstraction, not patient understanding. This makes post-visit coordination difficult, especially for aging adults or those with cognitive conditions. Caregivers are expected to piece together care across multiple systems, despite not being formally included in care team structures. APIs and SMART on FHIR apps are technically available, but inconsistent implementation and data completeness make them unreliable for longitudinal patient insight.

### Opportunities

Patients should have the right to access their full clinical history—including structured and unstructured data—regardless of the system or vendor. This should include imaging, referrals, and progress notes. CMS can promote consistent support for standards-based interoperability through TEFCA and FHIR, but should also fund pilots for ambient, AI-supported longitudinal care navigation tools for caregivers.

New rules should require that caregiver access and shared decision-making features be standard in-patient portals and third-party app interfaces. AI development and deployment is moving faster than traditional development and adoption cycles. To advance usability and accessibility at scale, we must also move from protracted innovation timelines to agile, rapid development cycles.

### Example from Practice

As a caregiver, I routinely navigate a maze of patient portals and care silos to manage medications and lab results for multiple family members. Even as a health informaticist who built a SMART on FHIR app to retrieve my own health data, I struggled to connect records across systems. For most patients, this is simply unmanageable.

An AI agent I designed now integrates CGM and wearables data to provide personalized feedback and insight for self-management. Expanding access to comprehensive data through FHIR-enabled apps and QHINs would make this kind of real-time, personalized support possible for millions of patients and caregivers.

**Policy Recommendation**

CMS should require certified EHR vendors and QHINs to support a comprehensive patient record API that includes structured and unstructured data. These APIs should operate within a broader ecosystem that reflects not only technical capabilities but also cross-sector accountability and governance. This API must adhere to a 'Portal-API Data Parity' standard, ensuring that any data available to the patient or caregiver via a portal is equally accessible and consumable via the API. This API must include not only structured and unstructured clinical data (e.g., progress notes, imaging reports, referral details), but also patient-generated health data (PGHD) from wearables, patient-reported outcomes (PROs), and critical caregiver observations that are currently siloed or uncaptured (e.g., medication adherence challenges, subtle behavioral changes, social determinants of health insights).

While acknowledging potential resource constraints for smaller practices, CMS should fund and require AI governance frameworks at state and organizational levels... *with tiered requirements and accessible resources to ensure equitable adoption across all provider types.*"

## Digital Health Application Accessibility and User Experience

*This directly addresses RFI questions PC-3, PC-4, PC-6, and PC-14, which explore existing digital health applications, missing features, accessibility needs, and digital identity adoption.*

**Barriers**

Current digital health applications often fail to account for the diverse needs of Medicare beneficiaries, particularly those with limited digital literacy or accessibility requirements. Many applications prioritize younger, tech-savvy users while overlooking the specific workflow needs of family caregivers managing complex care coordination.

Digital identity systems remain fragmented and inconsistent across health systems, creating barriers to seamless data access. The complexity of managing multiple logins and authentication methods particularly burdens older adults and caregivers managing multiple family members' care.

**Opportunities**

CMS should establish accessibility standards for Medicare-focused digital health applications that go beyond basic Web Content Accessibility Guidelines (WCAG) compliance to address the specific needs of aging adults and family caregivers. This includes support for voice interfaces, computer vision, simplified navigation patterns, and caregiver proxy access capabilities.

Investment in standardized digital identity frameworks that work seamlessly across health systems would significantly reduce barriers to patient and caregiver engagement with digital health tools.

# III. Provider Responses

## Supporting Provider Adoption of Digital Health Innovation

*This directly addresses RFI questions PR-1, PR-2, PR-7, and PR-9, which focus on provider adoption strategies, workflow application obstacles, comprehensive data availability, and digital identity adoption.*

**Barriers**

Rapid adoption of AI tools is outpacing many health systems' readiness to evaluate, govern, and implement these technologies responsibly. There is proposed limiting of federal- state-level coordination, creating regulatory blind spots and duplicative efforts that mirror HIPAA-era misalignment. Community-based organizations and safety-net providers often lack resources to vet AI tools or build out governance infrastructure. In addition, there are limited provider training and tools to effectively capture and integrate caregiver-provided data or to seamlessly coordinate care with family caregivers, leading to fragmented information flow. Many AI tools are marketed based on features rather than actual function, performance, or clinical utility.

**Opportunities**

CMS should promote the development of state-aligned AI oversight structures that are agile, responsive, and designed for local relevance, while adhering to federal guardrails.

CMS should fund and require AI governance frameworks at state and organizational levels with tiered requirements and accessible resources to ensure equitable adoption across all provider types, acknowledging potential resource constraints for smaller practices.

Rather than relying on traditional, slow-moving pilot programs, CMS can accelerate responsible AI deployment by modeling new grant structures after successful community mobilization frameworks—such as the Drug-Free Communities (DFC) Support Program. Due to the rapid development of foundational LLMs, a three-year,

community-driven funding model would enable local health coalitions—including Medicaid agencies, safety-net providers, standards organizations, and caregiver groups—to co-develop and test AI governance practices in real-world environments.

This approach encourages rapid-cycle learning and measurable outcomes with humans in the loop, without the inertia of legacy pilot structures, and would also help address implementation gaps by elevating practical, cross-sector solutions rooted in lived experience and operational needs.

Additionally, CMS should promote and fund provider-facing digital tools that facilitate secure, structured input from caregivers and integrate this 'invisible health data' into clinical workflows (e.g., validated caregiver-reported outcome measures, shared care plan platforms). AI tools used in care delivery—such as ambient listening, computer vision, clinical summarization, and patient monitoring—should also be validated in diverse settings, with special emphasis on Medicaid and dual-eligible populations to ensure equitable outcomes.

CMS should further strengthen this work by partnering with organizations already advancing AI governance and safety, including The Joint Commission and the Coalition for Health AI (CHAI), to codify, adapt, and disseminate practical governance guidance for AI algorithms that reflects both national standards and local realities.

**Example from Practice: The Imperative for Human Oversight in AI**

My daily work as an AI strategist and my personal experiences as a caregiver consistently reinforce the critical need for human oversight in AI-driven health solutions. For instance, in a recent demonstration using a large language model with computer vision to identify medications, the AI initially misidentified a common drug, Dicyclomine, and its purpose. It was only through direct human correction that the error was caught and rectified.

This experience, and many others, underscores how even advanced AI can exhibit "fragile performance" without robust governance frameworks and a human-in-the-loop design.

Such direct, real-world interactions inform my work with national organizations like NCPDP, payers, and vendors, helping them understand the complex intersection of AI governance, standards-based data exchange, and the absolute necessity of human verification.

**Policy Recommendation**

CMS should fund and require AI governance frameworks at both state and organizational levels. These frameworks should include clear expectations for

implementation transparency, data lineage, human oversight, and performance monitoring relevant to care coordination with families. A national registry of clinical AI tools, modeled after the Joint Commission–CHAI partnership, could provide reference implementation standards.

# IV. Technology Developer Responses

## Interoperability Standards and API Implementation

*This directly addresses RFI questions TD-1, TD-5, TD-7, TD-9, TD-13, and TD-15, which focus on developer incentives, FHIR endpoints directory, USCDI effectiveness, certification changes, complete EHI access, and bulk FHIR API usage.*

**Barriers**

Current API implementations lack consistency across vendors, making it difficult for developers to create reliable applications that work across different health systems. The absence of a comprehensive FHIR endpoints directory creates discovery barriers for both developers and patients seeking to connect their data.

API designs often overlook the specific data access and functionality needs of third-party applications designed for caregivers, such as bulk data access for longitudinal care management or proxy access features.

Certification requirements focus heavily on technical compliance rather than real-world usability and performance, leading to systems that meet regulatory requirements but fail to deliver meaningful value to patients and providers.

**Opportunities**

CMS should mandate and fund a national Provider API Directory that is as accessible and user-friendly as possible. This directory must include comprehensive metadata about available data types, authentication requirements, and real-world performance metrics, alongside clear documentation for developer access to caregiver-relevant data elements and proxy functionality.

Furthermore, CMS should promote API design principles that prioritize caregiver app interoperability within this directory.

Certification criteria should evolve to emphasize API usability, data completeness, and real-world performance rather than just technical compliance. This includes requiring "sandbox fidelity" where developer testing environments accurately reflect production system capabilities.

**Policy Recommendation**

Implement mandatory "Portal-API Data Parity" requirements ensuring that any data visible in patient portals is also available via certified FHIR APIs. Support dynamic client registration protocols and verifiable credentials to enable real-time, secure application authorization without lengthy pre-registration processes.

# V. Value-Based Care Responses

## AI Integration in Alternative Payment Models

*This directly addresses RFI questions VB-1, VB-2, VB-3, VB-11, and VB-14, which focus on APM digital health leverage, AI integration, critical health IT capabilities, interoperability challenges, and digital identity benefits.*

**Barriers**

Current value-based care models often lack the infrastructure to effectively capture and utilize the comprehensive data needed for population health management. Many APM participants struggle with data integration across multiple sources, limiting their ability to identify high-risk patients and coordinate care effectively.

**Opportunities**

AI-powered population health analytics and risk stratification integrating AI-powered population health analytics and risk stratification tools into APM requirements, with specific focus on incorporating social determinants of health and caregiver-reported outcomes and burden assessments. CMS should incentivize APM participants to adopt comprehensive data integration platforms that include wearable devices, patient-reported outcomes, and caregiver observations.

**Policy Recommendation**

CMS should create specific incentives for APM participants who demonstrate effective use of AI governance frameworks and comprehensive data integration. This includes recognition and payment adjustments for organizations that successfully incorporate invisible health data—such as caregiver coordination activities—into their care management processes.

# VI. Conclusion

AI and interoperability are no longer future-facing challenges—they are present-tense pressures reshaping care, cost, access, and trust. To ensure our health technology

ecosystem advances safely and equitably, we must ground innovation in the lived realities of patients, caregivers, and community-based providers—not just the limits of today's technical infrastructure.

Through my work as an AI strategist, FHIR implementer, and caregiver-advocate, I have witnessed the systemic gaps this RFI seeks to address. I have also seen how thoughtful policy, aligned governance, and collaboration across federal, state, and private sectors can build the scaffolding needed to translate emerging tools into long-term trust and impact.

I welcome the opportunity to share insights from Mosaic Life Tech's AI governance work, pilot implementation strategies shared with organizations, and the real-world lessons drawn from patients and caregivers navigating today's fragmented digital landscape.

If CMS convenes stakeholder groups, pilot programs, or implementation-focused tiger teams stemming from this RFI, I would be happy to contribute further—particularly around:

- Responsible AI implementation and governance frameworks
- Equity-centered design for APIs and caregiver technologies
- State-federal coordination for data, standards, and payment innovation

Thank you for the opportunity to provide this input.

Sincerely,

Teresa Younkin, MSHI
AI Strategist | Health Informaticist
Founder, Mosaic Life Tech, LLC